

Budapest University of Technology and Economics

Faculty of Natural Sciences

M3 - Study of Comparators

Lab report

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Lab responsible
Viktor Farkas

Student
Abigél ORTUTAY (UGHRKC)
Physics-engineering BSc 3rd year
abico.ortutay@gmail.com

Student
Hanna ERTL (SPH4U7)
Physics-engineering BSc 3rd year
hannavedaertl@gmail.com

Abstract

1 Theoretical introduction

2 Measurement tools

For this measurement, we used the following list of tools:

- ST μA741CN operational amplifier
- OK Industries Inc. CM-140 L breadboard
- Resistors: 22 kΩ (1 piece), 10 kΩ (2 pieces), 100 kΩ (1 piece), 1 kΩ (1 piece)
- Jumping wires (6 pieces)
- Crocodile clips (6 pieces)
- Banana plugs (6 pieces)
- TI TL071CP operational amplifier
- Capacitor: 6.8 nF (1 piece)
- HM8040-2 HAMEG Triple power supply
- HM 8011-3 HAMEG Digital multimeter
- SIGLENT SDS 1072CML Digital storage oscilloscope
- Handheld multimeter

3 Measurement setup

4 Measurements

4.1 Switching voltage of the comparator with an operational amplifier

For the first measurement, we had the ciruit setup as seen on figure 1 and described in the Measurement Setup section above.

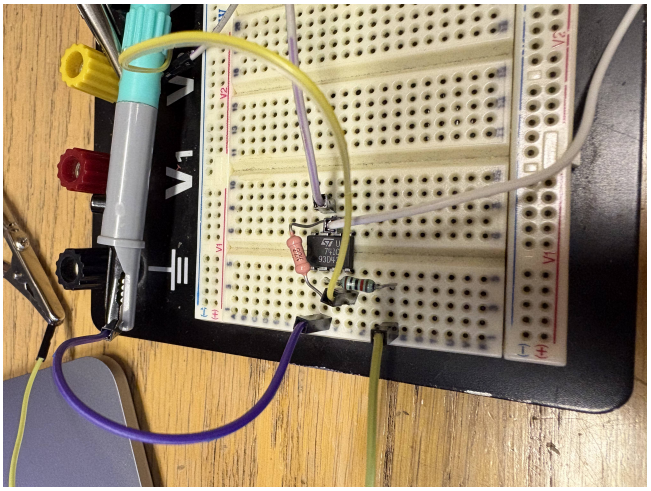


Figure 1: Simple comparator setup with two resistors

We wanted to determine the switching threshold of the configuration. We began varying the the voltage, satrting from 5V on the input, we found the output to be 4.468V. At this voltage we expected the output to be saturated to the positive supply value, which is mostly what we found, with a deviation of value:

$$Deviation = \frac{U_{+supp} - U_{out}}{U_{+supp}} = \frac{5 - 4.468}{5} = 0.106$$

As the configuration acts as a voltage divider on the non-inverting input of the operational amplifier, from this we could calculate that we suspect the switching threshold to be around:

$$U_+ = \frac{R_2}{R_1 + R_2} \cdot U_{+supp} = \frac{10}{32} \cdot 5 = 1.5625V$$

Varying the voltage, experimentally we found the switching threshold to be at 1.6 V, where the output showed -3.058 V on the multimeter.
While the measured switching voltage is close to the theoretical value we expected at only 2.4% deviation, we expected the output to saturate to the negative supply at -5 V. However, we found that the output went no lower than -3.058 V.
We believe this could be because the operational amplifiers we used were not rail-to-rail devies. In this case the internal structure of the amplifier is such that voltage drops occur on the internal output stage transistors. The asymmetry of the voltage drops in the positive and negative saturation regimes can be due to the asymmetric output stage on which this voltage

drop occurs.
The difference we found between the magnitude of the output voltages on the two regimes was close to 4.468 - 3.058 = 1.41 V.

4.2 New switching voltage in Schmitt-trigger with operational amplifier

Altering the setup of the previous task by adding a resistor of 100 kΩ, we created the Schmitt-trigger setup. We expected the switching threshold to extend to an interval, as the purpose of the Schmitt-trigger setup was to reduce the sensitivity to input noise. We calculated the theoretical switching thresholds to be at:

$$U_1 = \frac{5}{1 + \frac{R_1}{R_2} + \frac{R_1}{R_3}} + \frac{U_{\text{out}}}{1 + \frac{R_3}{R_2} + \frac{R_3}{R_1}} = \frac{5\text{ V}}{1 + 2.2 + 0.22} + \frac{U_{\text{out}}}{1 + 10 + 4.545} = 1.462\text{ V} + \frac{U_{\text{out}}}{15.545}$$

$$U_{1+} = 1.462\text{ V} + \frac{5\text{ V}}{15.545} = 1.462\text{ V} + 0.322\text{ V} = 1.78\text{ V}$$

$$U_{1-} = 1.462\text{ V} + \frac{-5\text{ V}}{15.545} = 1.462\text{ V} - 0.322\text{ V} = 1.14\text{ V}$$

The measured switching thresholds were at 1.8 V and 1.3 V, at these thresholds the output switched to -3.086 V and 4.36 V respectively. The higher threshold - below which the output voltage saturates to the negative supply voltage - is very close to the theoretical value, with only 1% deviation. The lower threshold - where the output saturated to the positive supply voltage - deviated by 14%. We believe this signification deviation could also be a result of the resolution of the measurement device used. We could only be precise to 1 decimal point, at which point the deviation could have also been 9% at 1.25 V on the input, rounded up to 1.3 V.
The difference between the output voltage magnitudes stayed relatively unchanged, compared to the previous task.

4.3 Waveforms of the relaxation oscillator with operational amplifier

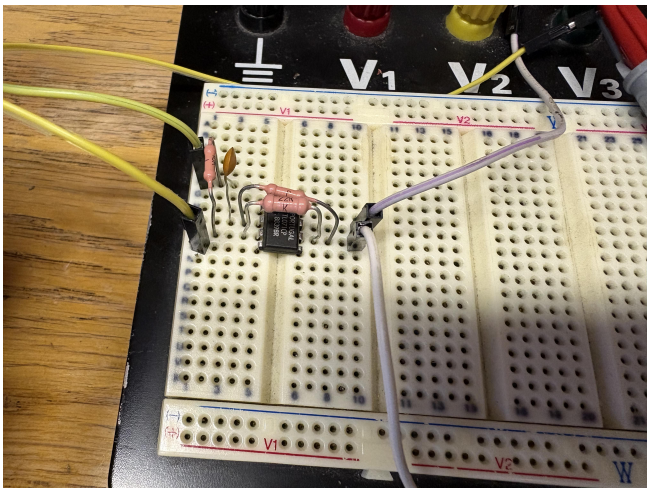
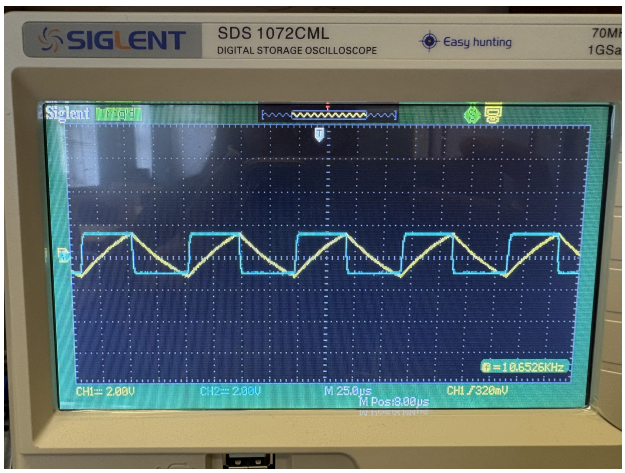
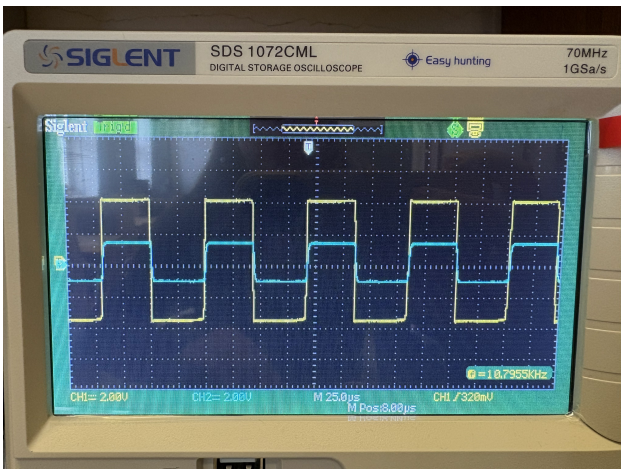


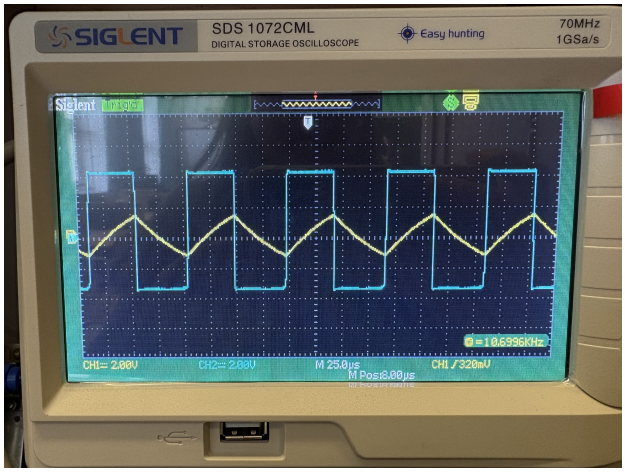
Figure 2: Relaxation oscillator setup with three resistors and one capacitor



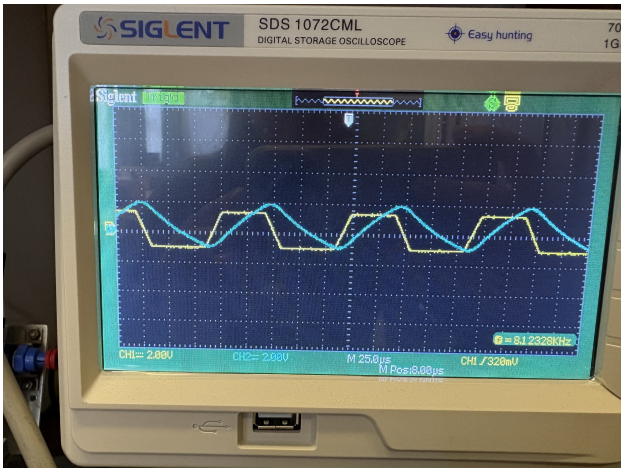
(a) Measurement of U_1 on channel 2 and U_2 on channel 1



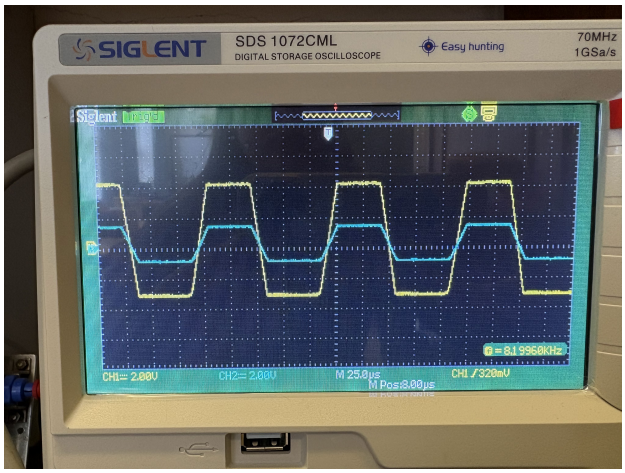
(b) Measurement of U_1 on channel 2 and U_{out} on channel 1



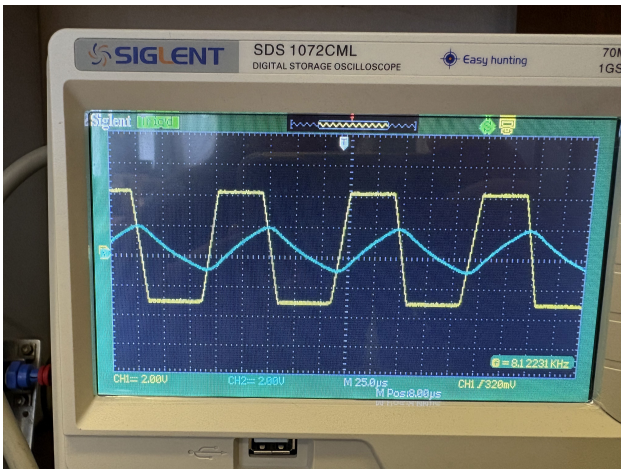
(c) Measurement of U_{out} on channel 2 and U_2 on channel 1



(d) Measurement of U_2 on channel 2 and U_1 on channel 1



(e) Measurement of U_1 on channel 2 and U_{out} on channel 1



(f) Measurement of U_2 on channel 2 and U_{out} on channel 1

Figure 3: Different measured waveforms of the relaxation oscillator setup with two operational amiplifiers, TL071CP and μA741CN; varying the electrodes measured on the channels of the oscilloscope.

4.4 Switching thresholds of the transistor Schmitt trigger circuit

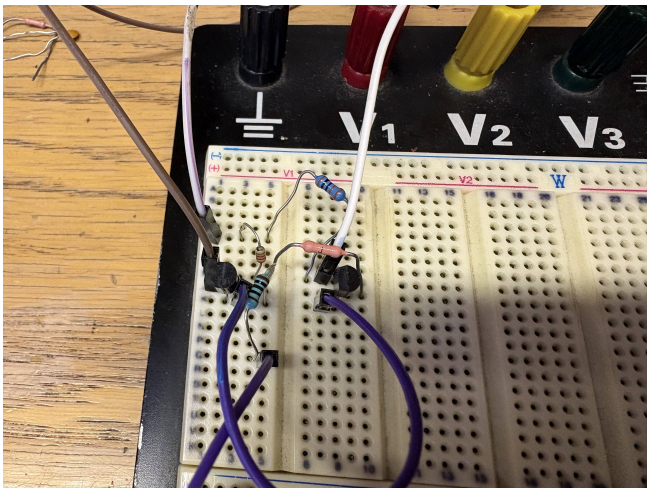


Figure 4: Schmitt trigger circuit setup with two transistors and four resistors

5 Conclusion